Printed Page:-04 **Subject Code:- ACSE0306 / ACSEH0306**

Roll. No:

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY, GREATER NOIDA

(An Autonomous Institute Affiliated to AKTU, Lucknow)

B.Tech

SEM: III - THEORY EXAMINATION DEC - 2023

Subject: Discrete Structures

Time: 3 Hours **Max. Marks: 100**

General Instructions:

IMP: *Verify that you have received the question paper with the correct course, code, branch etc.*

1. *This Question paper comprises of* **three Sections -A, B, & C.** *It consists of Multiple Choice Questions (MCQ's) & Subjective type questions.*

2. *Maximum marks for each question are indicated on right -hand side of each question.*

3. *Illustrate your answers with neat sketches wherever necessary.*

4. *Assume suitable data if necessary.*

5. *Preferably, write the answers in sequential order.*

6. *No sheet should be left blank. Any written material after a blank sheet will not be evaluated/checked.*

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SECTION A **20**

1. Attempt all parts:-

1-a. The set of positive integers is _____. (CO1) 1

(a) Infinite

(b) Finite

(c) Subset

(d) Empty

1-b. A={a,e,i,o,u} is an example of which of the following (CO1) 1

(a) Set Builder Notation

(b) Roster Form

(c) Both of the mentioned options

(d) None of the mentioned options

1-c. In a group there must be only _____ element.(CO2) 1

(a) 2

(b) 1

(c) 3

(d) 4

1-d. A trivial subgroup consists of _____ (CO2) 1

(a) Cosets

(b) Inverse element

(c) Ring

(d) Identity element

1-e. The logic gate that provides high output for same inputs _____ (CO3) 1

(a) NOT

(b) X-NOR

(c) AND

(d) XOR

1-f. The max term when X=Y=Z=1 is _____ (CO3) 1

(a) x'+y'+z'

(b) x'+y+z

(c) y'+z'

(d) x+y+z

1-g. If A is any statement, then which of the following is not a contradiction? (CO4) 1

(a) A ∧ ¬A

(b) A V F

(c) A ∧ F

(d) None of mentioned

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1-h. Let P: Today is cloudy day, Q: We should go for swimming. Then 'Today is cloudy day and we should not go for swimming.' is best represented by (CO4) 1

(a) P V Q

(b) ~P V Q

(c) P ∧ Q

(d) P ∧ ~ Q

1-i. Choose the traversal outputs the data in sorted order in a BST (CO5) 1

(a) Preorder

(b) Inorder

(c) post order

(d) Level order

1-j. A graph is Planer graph if and only if it does not contain a subgraph 1

homeomorphic to k5 or k3,3.(CO5)

(a) TRUE

(b) FALSE

(c) all the above

(d) none

2. Attempt all parts:-

2.a.	Find the power set of S where S={ 1, 2, 3}. (CO1)	2
2.b.	Show that, the set of all integers is a group with respect to addition.(CO2)	2
2.c.	Draw the Hasse diagram for the relation divisibility of (S, /) where S={1,2,3,4,6,8,12}. (CO3)	2
2.d.	Write rules of inference are used in this famous argument. "All men are mortal. Socrates is a man. Therefore, Socrates is mortal." (CO4)	2
2.e.	Explain in brief graph coloring with a suitable example.(CO5)	2

SECTION B **30**

3. Answer any <u>five</u> of the following:-

3-a.	Prove that: $A \cap B = B \cap A$. (CO1)	6
3-b.	Consider relation R on set A ={ 7, 8, 9} depicted by R ={ (7,8), (7,9), (8,8), (9,7)}. Find the reflexive closure of R. (CO1)	6
3-c.	State and prove the Lagrange's Theorem. (CO2)	6
3-d.	Differentiate among Monoid, Semi Group, Group and Abelian group with an example. (CO2)	6
3.e.	Define about a relation R that is both a partial ordering and on equivalence relation. Give an example of such relation on the set {1,2,3}. (CO3)	6
3.f.	Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent. (CO4)	6
3.g.	Write short note on : a) Homomorphism of graphs b) Euler and Hamiltonian Paths (CO5)	6

SECTION C **50**

4. Answer any <u>one</u> of the following:-

4-a.	State two ways to represent the relation with an example. Differentiate among symmetric, asymmetric and antisymmetric relation. (CO1)	10
4-b.	A= {1, 2, 3} R= { (1, 2), { 2, 3), (3, 1). Find following for R, a) Symmetric closure b) Reflexive Closure c) Transitive Closure (CO1)	10

5. Answer any <u>one</u> of the following:-

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5-a. Prove that $(Z_6, (+_6))$ is an abelian group of order 6, where $Z_6 = \{0,1,2,3,4,5\}$. (CO2) 10

5-b. Prove that, If in a group (G,*), 'x', 'y' and 'z' are three elements such that x * y = z * y, then x = z. (CO2) 10

6. Answer any <u>one</u> of the following:-

6-a. Show that with an example that the union of two sublattice may or may not be a sub lattice.(CO3) 10

6-b. A logic circuit is to be constructed that will implement the Boolean expression: X = A B' C + A' B C+A'B'C' (CO3) 10

7. Answer any <u>one</u> of the following:-

7-a. Let F (x, y) be the statement "x can fool y," where the domain consists of all people in the world. Use quantifiers to express each of these statements. (CO4) 10

a) Everybody can fool Fred.

b) Evelyn can fool everybody.

c) Everybody can fool somebody.

d) There is no one who can fool everybody.

e) Everyone can be fooled by somebody.

7-b. Translate these statements into English. (CO4) 10

a) $\forall x \exists y$ love(x, y)

b) $\forall x$ (person(x) $\rightarrow$ love (Mary, x))

c) $\forall x$ ((student(x) & love (x, Mary)) $\rightarrow$ happy (x)))

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8. Answer any <u>one</u> of the following:-

8-a. Define closed walk, open walk, path and circuit. Take a graph of your choice and give an example to each one. (CO5) 10

8-b. Prove that a connected graph with n vertices must have n-1 edges. Can a single undirected graph of 8 vertices have 40 edges excluding self loop. (CO5) 10